Supplementary material

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S1. Run-level tests

Table S1 lists every comparison in the paper. Households scored by one trained model are not independent, so the unit of replication is the training run: each run contributes its pooled test MAE. Where both arms ran with the same seeds, the differences were paired by seed (paired t-test); otherwise Welch's t-test was used. With every household present in every run, this is equivalent to testing the arm in a linear mixed model with crossed random effects for household and run. The Holm correction was applied over all rows together, and it is the only correction used in the paper. ΔMAE is A minus B; negative values favour A. The share of households doing better under A is descriptive only.

Table S1. Run-level comparisons (10^{-4} kWh, 95 % confidence interval). Test: paired by seed (n pairs) or Welch ($n_A + n_B$ runs).

A	B	Test (runs)	ΔMAE A−B (10^{-4} kWh) [95% CI]	Homes A better (%)	p_{Holm}
Centralised	Local-only	paired, n=5	−27.0 [−33.8, −20.2]	72	0.011
FedAvg, K=1	Centralised	paired, n=5	−2.9 [−8.5, +2.8]	49	1.000
FedAvg, K=20	Centralised	paired, n=5	+41.9 [+30.2, +53.6]	1	0.016
FedAvg, K=100	Centralised	paired, n=5	+80.9 [+69.2, +92.5]	1	0.0013
Windows, K=20	Homes, K=20	paired, n=5	−6.1 [−16.0, +3.7]	71	1.000
Windows, K=100	Homes, K=100	paired, n=5	−18.9 [−34.5, −3.2]	65	0.598
FedAvg, cluster	FedAvg, random	paired, n=5	+2.7 [−4.3, +9.6]	44	1.000
FedAvg, shape	FedAvg, random	paired, n=5	+2.7 [−6.5, +11.8]	51	1.000
FedProx, α=0.3	FedAvg, α=0.3	paired, n=5	−2.5 [−5.3, +0.4]	79	1.000
HA-FedProx, cluster	FedAvg, cluster	paired, n=5	+5.7 [+0.7, +10.7]	27	0.682
FedAvg, α=0.3, 16-bit	FedAvg, α=0.3	paired, n=5	−2.4 [−5.4, +0.6]	92	1.000
FedAvg, α=0.3, C=0.5	FedAvg, α=0.3	paired, n=5	−9.3 [−18.2, −0.5]	94	0.807
FedAvg, K=5	Client-only, K=5	paired, n=5	−15.2 [−33.0, +2.6]	63	1.000
FedAvg, K=20	Client-only, K=20	paired, n=5	+3.1 [−5.0, +11.3]	46	1.000
FedAvg + FT, K=5	Client-only + FT, K=5	paired, n=5	−20.1 [−30.1, −10.1]	91	0.120
FedAvg + FT, K=20	Client-only + FT, K=20	paired, n=5	−18.8 [−26.1, −11.4]	83	0.054
FedAvg + FT, α=0.3	Local-only	paired, n=5	−6.2 [−9.1, −3.3]	50	0.103
FedAvg, K=100	Local-only	paired, n=5	+53.8 [+40.1, +67.5]	18	0.012
FedAvg + FT, K=100	Local-only	paired, n=5	+10.0 [+3.0, +17.1]	41	0.372
Windows, K=20, plateau	Homes, K=20, plateau	paired, n=3	−3.8 [−17.4, +9.9]	44	1.000
Windows, K=50, plateau	Homes, K=50, plateau	paired, n=3	−20.4 [−49.7, +8.9]	63	1.000
Windows, K=100, plateau	Homes, K=100, plateau	paired, n=3	−15.2 [−38.8, +8.4]	48	1.000
Windows, K=20, plateau	Centralised	paired, n=3	+11.9 [−4.4, +28.1]	9	1.000
Windows, K=100, plateau	Centralised	paired, n=3	+29.7 [−0.0, +59.5]	2	0.878
Homes, K=100, plateau	Centralised	paired, n=3	+44.9 [+27.7, +62.1]	6	0.180
Adam kept, K=20	Adam reset, K=20	paired, n=3	+6.0 [−16.1, +28.1]	18	1.000
Adam kept, K=100	Adam reset, K=100	paired, n=3	+0.8 [−18.0, +19.6]	35	1.000
SGD, K=20	Adam reset, K=20	paired, n=3	+47.1 [+39.9, +54.3]	10	0.034
SGD, K=100	Adam reset, K=100	paired, n=3	+25.6 [+0.3, +50.9]	19	0.878
FedAdam, K=20	Adam reset, K=20	paired, n=3	+10.0 [−1.3, +21.3]	48	0.995
FedAdam, K=100	Adam reset, K=100	paired, n=3	−16.3 [−42.9, +10.2]	65	1.000

S2. Training budget and round selection

Table S2 gives, for each number of clients, the local Adam steps per client and round, the round chosen on validation error and the test MAE, for the 30-round budget of the main text and for runs of up to 100 rounds that stopped after 25 rounds without a new best validation MAE; for the latter, the best and last round of each seed. Fig. S1 shows the validation curves. These 100-round runs are the reference for the optimiser comparison (Section S6); the plateau runs are in Section S5.

Table S2. Budget per number of clients K (FedAvg, random grouping). Steps/round: Adam steps per client per round. Best rd.: round selected on validation MAE (mean over seeds). Best / run: best round and rounds run per seed. Gap: test MAE relative to centralised.

K	Steps/round	Best rd. (30)	MAE (30)	Gap (%)	Best / run per seed (100)	MAE (100)	Gap (%)
1	4,535	18.6	0.0809	−0.4	–	–	–
5	907	21.8	0.0826	+1.8	–	–	–
20	227	22.0	0.0854	+5.2	91, 98, 100 / 100, 100, 100	0.0832	+2.5
50	91	26.6	0.0870	+7.2	52, 90, 25 / 77, 100, 50	0.0858	+5.7
100	46	29.2	0.0893	+10.0	99, 95, 92 / 100, 100, 100	0.0871	+7.3

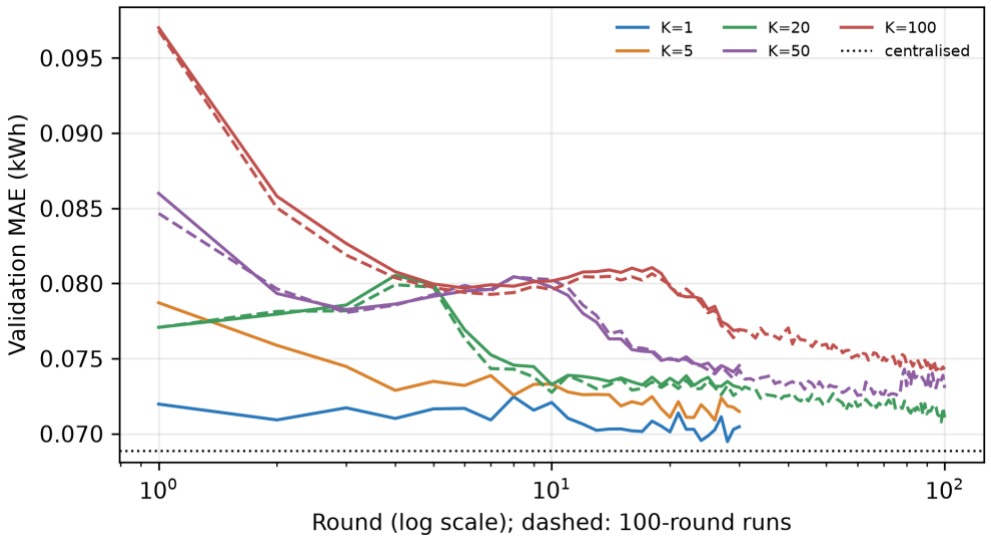


Figure S1. Validation MAE per round for each number of clients (mean over seeds). Solid: 30-round runs; dashed: runs of up to 100 rounds. Dotted: centralised.

S3. Heterogeneity-adaptive proximal coefficient

A rule was also tested that sets the proximal coefficient from the disagreement between client updates. After round r the server holds the updates $\Delta_k = w_k - w_r$ and their weighted mean $\overline{\Delta}$, and computes

$$\rho_r = \frac{\sum_k p_k \|\Delta_k - \overline{\Delta}\|^2}{\sum_k p_k \|\Delta_k\|^2} \in [0,1], \mu_{r+1} = \mathrm{clip}(\mu_0 \rho_r,\ 10^{-4},\ 1) \qquad \text{(S1)}$$

The rule never improved on FedAvg. Under the cluster partition its run-level difference from FedAvg was $\Delta = +5.7 \times 10^{-4}$ kWh (95 % CI +0.7 to +10.7, $p_{\mathrm{Holm}} = 0.682$), and 27 % of households did better under it. Its signal ρ_r ranked partitions by heterogeneity (Spearman +0.51 with H_{JS}, $p = 0.0044$, $n = 30$ runs), but on average it rose from 0.15 in round 1 to 0.96 in the last round, a change about 46 times its spread across partitions. As the shared descent direction shrinks during training, any residual disagreement pushes ρ_r towards one, so a coefficient proportional to it grows late in training whatever the data. Table S3 gives all three aggregation rules for each grouping.

Table S3. All aggregation rules for each grouping at $K = 20$ (mean $\pm$ s.d. over three to five seeds). $H = H_{\mathrm{JS}} + H_{\mathrm{CV}}/(1 + H_{\mathrm{CV}})$, a composite shown for reference; $\bar{\rho}$: mean update disagreement.

Partition	Method	H	H_{JS}	$\bar{\rho}$	Test MAE	Gap to central (%)
α=0.1	FedAvg	0.241	0.0189	0.874	0.0856 ± 0.0004	+5.4
α=0.1	FedProx	0.242	0.0181	0.863	0.0856 ± 0.0006	+5.4
α=0.1	HA-FedProx	0.242	0.0181	0.844	0.0859 ± 0.0003	+5.7
α=0.3	HA-FedProx	0.243	0.0153	0.844	0.0863 ± 0.0003	+6.3
α=0.3	FedAvg	0.246	0.0152	0.871	0.0860 ± 0.0003	+5.9
α=0.3	FedProx	0.246	0.0152	0.859	0.0857 ± 0.0002	+5.6
random	FedAvg	0.253	0.0106	0.858	0.0854 ± 0.0008	+5.2
random	FedProx	0.255	0.0105	0.845	0.0855 ± 0.0003	+5.3
random	HA-FedProx	0.255	0.0105	0.822	0.0859 ± 0.0009	+5.8
α=1.0	FedAvg	0.262	0.0125	0.868	0.0851 ± 0.0007	+4.8
α=1.0	FedProx	0.270	0.0124	0.856	0.0857 ± 0.0001	+5.6
α=1.0	HA-FedProx	0.270	0.0124	0.841	0.0857 ± 0.0008	+5.6
cluster	FedProx	0.381	0.0202	0.866	0.0858 ± 0.0002	+5.6
cluster	FedAvg	0.381	0.0202	0.876	0.0857 ± 0.0003	+5.5
cluster	HA-FedProx	0.381	0.0202	0.849	0.0862 ± 0.0004	+6.2

S4. Proximal coefficient grid

Table S4 gives the best validation MAE for each coefficient (cluster partition, seed 42). The row $\mu = 0$ is plain FedAvg.

Table S4. Best-round validation MAE (kWh) over the proximal coefficient grid.

μ (FedProx) or μ_0 (adaptive)	FedProx	Adaptive (HA-FedProx)
0	0.07238	0.07238
0.0001	0.07260	0.07260
0.001	0.07259	0.07268
0.01	0.07320	0.07332
0.1	0.07666	0.07682
1	–	0.07803

S5. Training to a plateau

FedAvg with homes grouped at random and with training windows dealt at random was trained for up to 500 rounds and stopped after 50 rounds without a new best validation MAE (three seeds). Table S5 gives, per seed, the round selected on validation and the number of rounds run, with the traffic up to the best round and in total. A run that reached 500 rounds was still improving, and its gap is an upper bound.

Table S5. FedAvg trained to a plateau. Clients: homes grouped at random, or training windows dealt at random (IID control). MB to best: traffic up to the selected round; MB total: traffic of the whole run (means over seeds).

K	Clients	Seeds	Best round (per seed)	Rounds run	Test MAE	Gap (%)	MB to best	MB total
20	homes	3	192, 98, 123	242, 148, 173	0.0826 ± 0.0007	+1.8	379	517
20	windows	3	261, 75, 66	311, 125, 116	0.0823 ± 0.0006	+1.3	369	507
50	homes	3	52, 113, 274	102, 163, 324	0.0847 ± 0.0008	+4.4	1,008	1,352
50	windows	3	346, 247, 154	396, 297, 204	0.0827 ± 0.0005	+1.9	1,715	2,059
100	homes	3	477, 202, 224	500, 252, 274	0.0856 ± 0.0008	+5.4	4,146	4,711
100	windows	3	295, 180, 53	345, 230, 103	0.0840 ± 0.0010	+3.5	2,424	3,113

S6. Optimiser set-up

At $K = 20$ (random grouping) and $K = 100$ (one home per client), FedAvg with Adam restarted every round was compared with client Adam whose state is kept across rounds, plain client SGD, and FedAdam, all for up to 100 rounds and stopped after 25 rounds without improvement (Table S6). Early rise: the largest rise of validation MAE above its running minimum in the first ten rounds. The SGD and server step sizes were chosen on validation MAE at $K = 20$ (seed 42) over the grid of Table S7.

Table S6. Optimiser set-up (FedAvg, up to 100 rounds; mean over seeds). Gap: test MAE relative to centralised.

K	Optimiser	Seeds	Best round	Test MAE	Gap (%)	Early rise (%)
20	Adam, reset each round	3	96	0.0832 ± 0.0002	+2.5	4.7
20	Adam, state kept	3	62	0.0838 ± 0.0007	+3.2	5.1
20	SGD	3	7	0.0879 ± 0.0001	+8.3	4.0
20	FedAdam (server)	3	23	0.0842 ± 0.0004	+3.7	1.8
20	FedAdam, step 0.03 (not selected)	3	30	0.0829 ± 0.0002	+2.1	27.1
100	Adam, reset each round	3	95	0.0871 ± 0.0004	+7.3	1.0
100	Adam, state kept	3	71	0.0872 ± 0.0005	+7.4	1.7
100	SGD	3	22	0.0897 ± 0.0007	+10.4	2.3
100	FedAdam (server)	3	35	0.0855 ± 0.0007	+5.3	24.7
100	FedAdam, step 0.03 (not selected)	3	30	0.0846 ± 0.0003	+4.2	27.9

Table S7. Step-size grid (validation MAE, $K = 20$ random grouping, seed 42, up to 100 rounds). Diverged: validation error not finite.

Optimiser	Step size	Best validation MAE (kWh)
FedAdam (server)	0.001	0.07516
FedAdam (server)	0.003	0.07193
FedAdam (server)	0.01	0.06954
FedAdam (server)	0.03	0.06963
Client SGD	0.1	0.07630
Client SGD	0.3	0.07551
Client SGD	1	diverged

S7. Error by consumption quintile

Households were divided into quintiles of mean training load (Q1 lowest). Table S8 gives the median household test MAE in each quintile, Fig. S2 the distribution of the benefit of each arm over local training, and Table S9 summary statistics of that benefit.

Table S8. Median household test MAE (kWh) by consumption quintile. FedAvg at $K = 20$ unless stated.

Quintile	Local-only	Centralised	FedAvg, α=0.3	FedAvg, α=0.3 + FT	FedAvg, K=100
Q1	0.0280	0.0285	0.0299	0.0297	0.0309
Q2	0.0476	0.0474	0.0494	0.0481	0.0513
Q3	0.0752	0.0741	0.0760	0.0782	0.0787
Q4	0.1058	0.1038	0.1099	0.1073	0.1118
Q5	0.1396	0.1319	0.1408	0.1393	0.1470

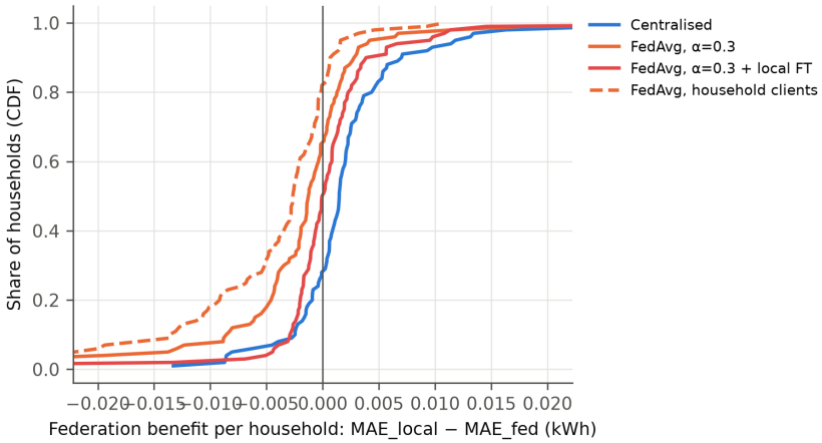


Figure S2. Distribution over households of the benefit of each collaborative arm over local training ($MAE_{local} - MAE_{arm}$, kWh; left of zero means worse off).

Table S9. Equity relative to local-only training (seed-averaged household MAE): share harmed (any difference), median benefit, 90th-percentile household MAE and share harmed after fine-tuning (FT).

Arm	Harmed vs local (%)	Median benefit (kWh)	P90 MAE	Harmed after FT (%)
Centralised	28	+0.0015	0.1362	–
FedAvg, random	61	−0.0008	0.1436	50
FedAvg, α=0.3	65	−0.0012	0.1442	50
FedAvg, cluster	63	−0.0008	0.1431	48
HA-FedProx, α=0.3	67	−0.0013	0.1454	55
FedAvg, K=100	82	−0.0026	0.1494	59
HA-FedProx, K=100	84	−0.0030	0.1515	65

S8. Calendar inputs and scaled error

Sine and cosine encodings of time of day and day of week were added as input channels to every arm in Table S10, with all other settings unchanged.

Table S10. Effect of calendar inputs, and MASE (scaled per household by the in-sample one-step naive MAE). Gain: MAE reduction relative to persistence; cal.: with calendar inputs. Gap: MAE relative to centralised training with the same inputs.

Arm	MAE	MASE	Gain (%)	MAE, cal.	MASE, cal.	Gain, cal. (%)	Gap, cal. (%)
Centralised	0.0812	0.989	+8.7	0.0815	0.996	+8.4	–
Local-only	0.0839	1.006	+5.7	0.0838	1.007	+5.8	–
FedAvg, K=20	0.0854	1.045	+4.0	0.0863	1.057	+3.1	+5.8
FedAvg, K=100	0.0893	1.109	−0.3	0.0919	1.158	−3.3	+12.8